

Learning dexterous robot hand control by imitating human hands

1st Yashuai Yan
Technische Universität Wien
Vienna, Austria
yashuai.yan@tuwien.ac.at

2nd Dongheui Lee
Technische Universität Wien
Vienna, Austria
dongheui.lee@tuwien.ac.at

Abstract—This paper presents an unsupervised deep-learning method for controlling dexterous robotic hands by mimicking human hand motions. We introduce a cross-domain similarity metric to capture the spatial and kinematic relationships between human and robot hands. Using this metric, our approach learns a shared latent space that aligns motion features across the two embodiments. The framework consists of two separate encoders that map human and robot hand data into the latent space, along with a robot decoder that generates feasible robot hand motions. During inference, only the human hand encoder and the robot hand decoder are needed to seamlessly retarget human hand movements to the robot hand, enabling scalable and flexible motion retargeting without requiring paired human-robot data. To demonstrate real-world applicability, we integrate our motion retargeting system with Mediapipe, a human hand pose estimator, enabling real-time robotic hand control from RGB video input.

Index Terms—motion retargeting, shared latent space, contrastive learning, dexterous robot hand.

I. INTRODUCTION

Multi-fingered robotic hands have the potential to replicate human hand movements, unlocking new possibilities across various domains of robotics, including dexterous manipulation, teleoperation, and human-robot interaction. However, designing an effective controller for these systems remains a significant challenge due to their high degrees of freedom (DoF) and the complexity of contact-rich tasks, especially when interacting with irregularly shaped objects.

Recent advancements in deep reinforcement learning (DRL) have shown promise in addressing these challenges. For example, [1] demonstrated

dexterous manipulation tasks, such as hammering and door opening, with a 24-DoF robotic hand. Similarly, [2] trained a vision-based DRL controller to manipulate a Rubik's cube using a single robotic hand. Despite their success, these methods often struggle with generalization to unseen tasks, and the black-box nature of RL optimization can lead to unsafe or unexpected behaviors.

Learning from human demonstrations offers an intuitive alternative, given the kinematic and morphological similarities between human and robotic hands. Demonstrations are typically collected via teleoperation systems, which often require specialized and expensive equipment, such as Virtual Reality (VR) or Augmented Reality (AR) devices, and wired gloves [3]–[5]. These setups can limit both the flexibility and scalability of data collection. In contrast, vision-based teleoperation with standard RGB cameras provides a more cost-effective and scalable solution. Vision-based teleoperation involves mapping human hand motions to robot hand motions to control the robot. To enhance motion diversity and explore the retargeting space, recent research leverages large-scale human motion capture datasets [6]–[8], which offer high-quality, diverse motion data. In this work, we focus specifically on motion retargeting from the human hand to the robot hand, using datasets like FreiHAND [8], which sufficiently cover the necessary range of human hand motions.

To address the motion retargeting problem, we propose a deep learning method that constructs a shared latent space for human and robot hand motions. This latent space preserves spatial joint

relationships and maintains the resemblance between human and robot motions. We achieve this through a combination of a cross-domain similarity metric and multiple loss functions. First, we use contrastive learning with a triplet loss to structure the latent space, pulling similar motions closer and pushing dissimilar ones apart. Second, a reconstruction loss is applied to robot data to ensure feasible motion generation from the latent space. Finally, a cyclic latent consistency loss is introduced to ensure that robot motions faithfully mimic the corresponding human motions. Notably, our approach learns the retargeting mapping in an unsupervised manner, without requiring paired human and robot motion data.

In summary, our method bridges the gap between human and robot hand motions by learning a shared latent representation, enabling effective and scalable motion retargeting without the need for costly teleoperation setups or labor-intensive paired datasets. This approach opens the door to more adaptable and robust robotic hand control, with potential applications spanning teleoperation, assistive robotics, and beyond.

II. RELATED WORK

A. Dexterous Manipulation

Dexterous manipulation with robotic hands has been a long-standing research challenge, remaining one of the most complex control tasks in robotics due to the high DoF and intricate contact dynamics involved [9], [10]. Recent advancements in DRL have demonstrated promising results for complex in-hand manipulation tasks. For example, OpenAI's work on rotating a Rubik's cube [2] and Rajeswaran et al.'s exploration of tasks like hammering and door opening [1] have highlighted the potential of learning-based approaches. However, these methods often struggle with sample efficiency, requiring large amounts of training data. Additionally, the learned policies are typically task-specific, with limited generalizability to new or unseen tasks.

B. Motion Retargeting

Motion retargeting aims to transfer human motion skills to robotic systems while accounting for differences in embodiment and kinematic constraints. Traditional optimization-based methods

define explicit correspondences between human and robot joints — for instance, aligning a human shoulder with a robot's shoulder joint. These methods then solve inverse kinematics (IK) to minimize the positional or orientation error [11]–[13]. While effective, these approaches often sacrifice motion nuances for stability, resulting in overly smoothed or unnatural movements, especially in dynamic tasks [12]. Similarly, manual joint selection techniques, like those in [14], limit the range of transferable motions and require expert intervention.

More recently, learning-based approaches have emerged as a powerful alternative. These methods learn a shared latent space to map human motions to robot motions. For example, [15] constructed a latent space using annotated paired data, while [16] introduced an automatic data generation process to reduce manual annotation efforts. Building on this, [17] employed unsupervised contrastive learning to learn a shared latent space from unpaired human and robot motion data, guided by a cross-domain similarity metric.

However, most prior work focuses on whole-body motion retargeting, neglecting the intricate complexities of hand manipulation. In this paper, we extend the idea of unsupervised contrastive learning to the human hand and robot hand, defining a hand-specific similarity metric to guide latent space learning. This enables us to capture the nuanced spatial relationships and dexterous capabilities of human hand motions, facilitating more natural and adaptable robotic hand control.

III. METHODOLOGY

In this section, we describe our unsupervised learning method for human-to-robot hand imitation.

A. Problem formulation

We represent the posture of the human hand as a feature vector $\mathbf{h} \rightarrow \mathbb{R}^n$ and the posture of the robot hand as a feature vector $\mathbf{r} \rightarrow \mathbb{R}^m$. Thus, the retargeting from human hand motion to the robot hand can be formulated as a function f that maps human hand $\mathbf{h}$ to robot hand $\mathbf{r}$: $f : \mathbf{h} \rightarrow \mathbf{r}$. The function f should preserve the visual resemblance of the human hand posture when generating a corresponding robot hand posture, e.g., the bending of fingers and the distance of fingertips.

Usually, retargeting domain-specific data between different domains requires paired data $(\mathbf{h}_i, \mathbf{r}_i)$ that $\mathbf{h}_i$ and $\mathbf{r}_i$ represent the same posture of the human and robot hands, respectively. The retargeting function f can be learned by minimizing the distance between $f(h_i)$ and r_i . However, collecting such datasets can demand significant manual effort. To reduce the demanding effort from humans, we adopt an unsupervised learning method from prior work, ImitationNet [17]. ImitationNet addresses motion retargeting from the human to the robot (excluding the hands) by learning a shared latent space. Both human and robot motions are encoded in the shared space, and the space is learned by unsupervised contrastive learning to pull similar data closer and dissimilar data further away. A domain-agnostic similarity metric is applied when learning such a shared latent space.

B. Hand representation

Human hands are usually represented as their joint positions in the Euclidian space, denoted as $\mathbf{x}_h \in \mathbb{R}^{n \times 3}$ where n is the number of joints of human hands. However, the 3D positions are sensitive to the definition of the coordinate system such as its origin and hand scaling. ImitationNet [17] uses the rotation-based representation, but the human and robot still have to be in the same coordinate system. Considering these limits, we first define a hand feature extraction function $\phi(\cdot)$ that takes a hand posture $\mathbf{x}_h$ as input and outputs a six-dimensional feature vector ($\phi(\mathbf{x}_h) = [b_1, b_2, b_3, b_4, b_5, \theta] \in \mathbb{R}^6$). The first five dimensions represent the flexion of five fingers, which are computed by

$$b_i = \frac{d_{root-tip}}{l_i}. \quad (1)$$

$d_{root-tip}$ is the distance between tip and root of the i -th finger, and l_i is the finger length. The last dimension, θ , denotes the additional adduction of the thumb finger.

The robot hand uses its joint angles to represent its posture. The advantage of joint angles is that the output of the retargeting function can be used to directly control the robot hand.

C. Similarity metric

Differing from the ImitationNet [17], which uses a rotation-based similarity metric, we define our similarity between the human hand and the robot hand based on our hand representation $\mathbf{h} = \phi(\mathbf{x}_h)$. Our hand feature includes the flexion of each finger and an additional adduction for the thumb. To obtain the same feature from the robot hand, we first compute its forward kinematic $\mathbf{x}_r = FK(\mathbf{r})$ to obtain 3D joint positions and then apply the same calculation as the human hand to obtain finger flexion and thumb adduction, which is denoted as $\phi(\mathbf{x}_r) \in \mathbb{R}^6$. The similarity score between the human hand and the robot hand is computed by

$$s = \|\phi(\mathbf{x}_h) - \phi(\mathbf{x}_r)\|_2. \quad (2)$$

The score s is used for contrastive learning when constructing the shared latent space.

D. Retargeting function and a shared space

Learning a cross-domain mapping function usually requires paired data from different domains. However, acquisition of such datasets might not be feasible. To tackle our hand imitation task, we apply unsupervised contrastive learning to construct a shared latent space between human and robot hands.

We employ three losses to learn the retargeting function and a shared latent space. Given a human hand dataset and robot hand dataset, we randomly select a batch of human and robot data and encode them to a shared latent space. From the encoded random variables, we randomly select three samples as a triplet. For instance, the three samples z^i, z^j and z^k are chosen, and then one sample is considered to be the anchor (e.g., z_o^i). The similarity scores of pairs (z^i, z^j) and (z^i, z^k) are computed by the metric defined in Section III-C. The negative and positive samples are determined by the scores (e.g., z_-^j, z_+^k). We use a triplet loss $\mathcal{L}_{triplet}$ to restrict the latent space so that the distance between z_o^i and z_+^k is at least smaller than the distance between z_o^i and z_-^j with a margin of $\alpha = 0.05$. The triplet loss is expressed as

$$\mathcal{L}_{triplet} = \max(\|z_o^i - z_+^k\| - \|z_o^i - z_-^j\| + \alpha, 0). \quad (3)$$

To generate feasible robot data from the latent space, we use a reconstruction loss between the

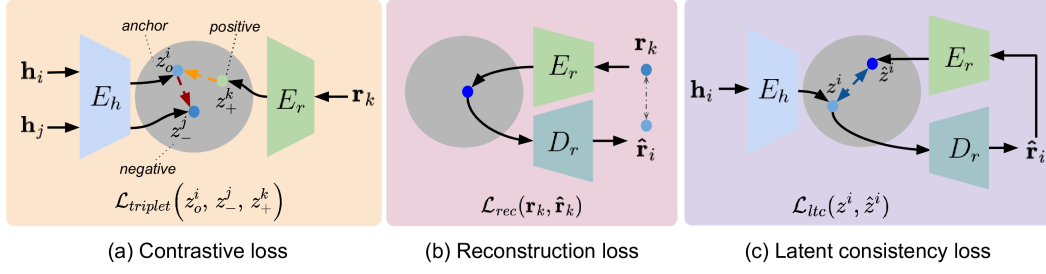


Fig. 1. **Training overview** we address the cross-domain retargeting by training three neural networks: a human hand encoder E_h , a robot hand encoder E_r and a robot hand decoder D_r . The encoders E_h and E_r encode random data from the human hand and robot hand to the shared space, respectively. Then, any three samples (e.g., z^i, z^j, z^k) are randomly chosen as triplets. Based on the defined similarity metric, the anchor z_o^i and the negative sample z_-^j and the positive sample z_+^k are decided. We use a contrastive loss to learn the shared space in (a). A reconstruction loss is imposed on the robot encoder and decoder to learn the robot motion reconstruction in (b). Since no paired data are acquired, our method learns to retarget the correct motion by ensuring latent consistency when encoding the generated hand motion into latent space through the robot encoder in (c).

input robot data $\mathbf{r}_k$ and the decoded robot data $\hat{\mathbf{r}}_k = D_r(E_r(\mathbf{r}_k))$. Our reconstruction loss is computed by

$$\mathcal{L}_{rec}(\mathbf{r}, \hat{\mathbf{r}}_k) = \|\mathbf{r} - \hat{\mathbf{r}}_k\|_2. \quad (4)$$

Finally, we apply a latent consistency loss $\mathcal{L}_{ltc}$ to guide the retargeting function, mapping human data to correct robot data. Due to the lack of paired data, there is no supervision to learn a direct mapping from $\mathbf{h}$ to $\mathbf{r}$. Our $\mathcal{L}_{ltc}$ loss ensures a cycle consistency between human latent $z^i = E_h(\mathbf{h})$ and the retargeted robot latent $\hat{z}^i = E_r(D_r(z^i))$. We calculate the loss as

$$\mathcal{L}_{ltc}(\mathbf{h}_i) = \|z^i - \hat{z}^i\|_2. \quad (5)$$

Our model learns the retargeting function f and the shared latent space by minimizing the three losses. We compute the weighted total loss by

$$\mathcal{L}_{total} = \lambda_1 \mathcal{L}_{triplet} + \lambda_2 \mathcal{L}_{rec} + \lambda_3 \mathcal{L}_{ltc}. \quad (6)$$

where we use $\lambda_1 = 10.0$, $\lambda_2 = 1.0$ and $\lambda_3 = 1.0$.

E. Inference control

After learning from random human and robot hand data, our decoder D_r is able to generate the robot hand from encoded human hand data while preserving a high similarity based on the metric in Section III-C. The generated robot hand data are joint angles that can be directly used to control the robot hand.

IV. EXPERIMENTS

In this section, we first describe our experimental setup and learning configurations when training and evaluating our method. Secondly, we assess the accuracy and fidelity of our model when retargeting human hand motions to the robot hand. Finally, we present our real-time hand imitation from RGB images.

A. Experimental setup

Our method trains three neural networks (see Fig. 1): human encoder E_h , robot encoder E_r , and robot decoder D_r . The networks are Multi-Layer Perceptrons (MLPs) with 6 linear layers, each with 256 units. The Exponential Linear Unit (ELU) [18] is used as the activation function, and the last layer is followed by a hyperbolic tangent function to normalize the output between -1 and +1. Our shared latent space has 32 dimensions. Adam [19] optimizer is utilized to minimize the total loss $\mathcal{L}_{total}$ in Eq. 6. The learning rate for the gradient descent step is 0.001. Our model is trained for around 0.5 hours on an RTX A4000 GPU and runs more than 1000 Hz on a CPU Intel i7-12700K.

We employ a tendon-driven robot hand RH8D that has two DoFs for the thumb and one DoF for the other fingers. Since the fingers have three joints but only one actuator, we set the three joints to the same angle in the simulation. We use the built-in controller to control the robot in the real world. The input to the controller is the joint

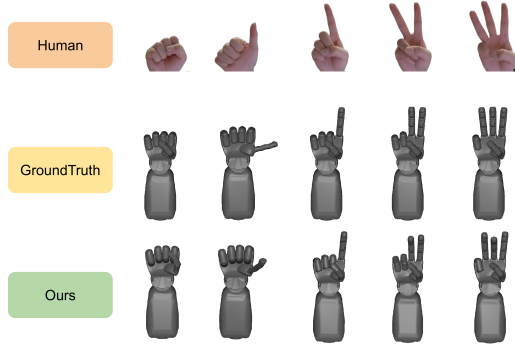


Fig. 2. **Hand retargeting in wild data:** We annotated several distinct human hand poses to evaluate our method. The retargeted robot hand poses are closely resembling the manually annotated ground-truth poses.

angles of the fingers, which is the output of our robot decoder.

B. Dataset

We employ the dataset FreiHAND [8] as human hand motion data. FreiHAND contains 32560 unique training samples and 3960 unique evaluation samples. Each data sample consists of an RGB image and a 3D hand pose. In this work, we use the 3D hand poses and preprocess them to our hand feature data (see Sec. III-B), and then split 80% of them for training and 20% for validation.

For the robot hand dataset, we randomly sample robot joint angles within their ranges and compute the FK online for each training step, rather than collecting a robot dataset. We employ PytorchKinematics [20] to parallelize the FK computation on the GPU to speed up the training.

C. Quantitative analysis

To evaluate the performance of our hand retargeting method, we annotate five distinct human hand poses, with the corresponding robot hand poses serving as ground-truth data. We used MediaPipe [21] to extract human hand poses from RGB images. As shown in Fig. 2, the first row presents the human hand poses used as test data, the second row displays the ground-truth robot hand poses, and the third row illustrates the retargeted robot hand poses.

We assess our model’s accuracy by calculating the mean squared error (MSE) between the annotated ground truth and the predicted robot



Fig. 3. **Irregular object grasping:** We grasp irregularly shaped objects in real-world scenarios by vision-based teleoperation. The grasping postures mimic the human hand grasping.

	Joint Angles (radin)	Control Frequency (kHz)
Ours	0.126	1.0

TABLE I

QUANTITATIVE ANALYSIS: WE COMPUTE THE MSE ERROR BETWEEN THE ANNOTATED JOINT ANGLE AND THE PREDICTED JOINT ANGLES BY OUR METHOD.

joint angles (see Table I). The retargeted robot hand accurately replicated the human hand poses, achieving an average error of 0.126 radians. This result demonstrates the method’s precise retargeting capability. Additionally, our approach can run at a high frequency, making it suitable for real-time and demanding applications.

D. Qualitative evaluation

We employ MediaPipe as the human hand pose estimator in the real world, and teleoperates the robotics hand to grasp different objects. Figure 3 shows the experiments carried out in wild scenarios. Thanks to our accurate human hand motion retargeting, our robotic hand successfully picks up irregular-shaped objects by imitating human grasping postures.

V. CONCLUSION

In this paper, we propose an unsupervised deep-learning approach for retargeting human hand motions to robotic hands. To alleviate the challenge of collecting paired data across different embodiments, we leverage contrastive learning to establish a shared latent space between human

and robot hands. We begin by defining a cross-embodiment similarity metric and then train separate encoders for the human and robot hands to map their poses into this shared space. A triplet loss encourages the network to bring similar hand poses closer together while pushing dissimilar ones apart. To ensure feasible robot motion generation, we incorporate a reconstruction loss, and a cyclic latent consistency loss reinforces faithful human-to-robot motion retargeting. Additionally, we implement vision-based teleoperation, enabling complex object grasping in unstructured, real-world environments.

REFERENCES

- [1] A. Rajeswaran, V. Kumar, A. Gupta, G. Vezzani, J. Schulman, E. Todorov, and S. Levine, "Learning complex dexterous manipulation with deep reinforcement learning and demonstrations," 2018. [Online]. Available: <https://arxiv.org/abs/1709.10087>
- [2] OpenAI, I. Akkaya, M. Andrychowicz, M. Chociej, M. Litwin, B. McGrew, A. Petron, A. Paino, M. Plappert, G. Powell, R. Ribas, J. Schneider, N. Tezak, J. Tworek, P. Welinder, L. Weng, Q. Yuan, W. Zaremba, and L. Zhang, "Solving rubik's cube with a robot hand," 2019. [Online]. Available: <https://arxiv.org/abs/1910.07113>
- [3] V. Kumar and E. Todorov, "Mujoco haptix: A virtual reality system for hand manipulation," in *2015 IEEE-RAS 15th International Conference on Humanoid Robots (Humanoids)*, 2015, pp. 657–663.
- [4] H. Hedayati, M. Walker, and D. Szafir, "Improving collocated robot teleoperation with augmented reality," in *2018 13th ACM/IEEE International Conference on Human-Robot Interaction (HRI)*, 2018, pp. 78–86.
- [5] Y. Pan, C. Chen, D. Li, Z. Zhao, and J. Hong, "Augmented reality-based robot teleoperation system using rgb-d imaging and attitude teaching device," *Robotics and Computer-Integrated Manufacturing*, vol. 71, p. 102167, 2021.
- [6] C. Ionescu, D. Papava, V. Olaru, and C. Sminchisescu, "Human3.6m: Large scale datasets and predictive methods for 3d human sensing in natural environments," *IEEE Transactions on Pattern Analysis and Machine Intelligence*, vol. 36, no. 7, pp. 1325–1339, 2014.
- [7] C. Guo, S. Zou, X. Zuo, S. Wang, W. Ji, X. Li, and L. Cheng, "Generating diverse and natural 3d human motions from text," in *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR)*, June 2022, pp. 5152–5161.
- [8] Z. Christian, C. Duygu, Y. Jimei, R. Bryan, A. Max, and B. Thomas, "Freihand: A dataset for markerless capture of hand pose and shape from single rgb images," in *IEEE International Conference on Computer Vision (ICCV)*, 2019. [Online]. Available: <https://lmb.informatik.uni-freiburg.de/projects/freihand/>
- [9] D. Rus, "In-hand dexterous manipulation of piecewise-smooth 3-d objects," *The International Journal of Robotics Research*, vol. 18, no. 4, pp. 355–381, 1999.
- [10] Y. Bai and C. K. Liu, "Dexterous manipulation using both palm and fingers," in *2014 IEEE International Conference on Robotics and Automation (ICRA)*, 2014, pp. 1560–1565.
- [11] K. Hu, C. Ott, and D. Lee, "Online human walking imitation in task and joint space based on quadratic programming," in *2014 IEEE International Conference on Robotics and Automation (ICRA)*, 2014, pp. 3458–3464.
- [12] L. Penco, B. Clement, V. Modugno, E. Mingo Hoffman, G. Nava, D. Pucci, N. G. Tsagarakis, J. B. Mouret, and S. Ivaldi, "Robust real-time whole-body motion retargeting from human to humanoid," in *2018 IEEE-RAS 18th International Conference on Humanoid Robots (Humanoids)*, 2018, pp. 425–432.
- [13] K. Darvish, Y. Tirupachuri, G. Romualdi, L. Rapetti, D. Ferigo, F. J. A. Chavez, and D. Pucci, "Whole-body geometric retargeting for humanoid robots," in *2019 IEEE-RAS 19th International Conference on Humanoid Robots (Humanoids)*, 2019, pp. 679–686.
- [14] S. Choi and J. Kim, "Towards a natural motion generator: a pipeline to control a humanoid based on motion data," in *2019 IEEE/RSJ International Conference on Intelligent Robots and Systems (IROS)*, 2019, pp. 4373–4380.
- [15] S. Choi, M. Pan, and J. Kim, "Nonparametric motion retargeting for humanoid robots on shared latent space," in *Robotics*, ser. Robotics: Science and Systems, M. Toussaint, A. Bicchi, and T. Hermans, Eds. United States: MIT Press Journals, 2020, publisher Copyright: © 2020, MIT Press Journals. All rights reserved.; 16th Robotics: Science and Systems, RSS 2020 ; Conference date: 12-07-2020 Through 16-07-2020.
- [16] S. Choi, M. J. Song, H. Ahn, and J. Kim, "Self-supervised motion retargeting with safety guarantee," 2021. [Online]. Available: <https://arxiv.org/abs/2103.06447>
- [17] Y. Yan, E. V. Mascaro, and D. Lee, "Imitationnet: Unsupervised human-to-robot motion retargeting via shared latent space," 2024. [Online]. Available: <https://arxiv.org/abs/2309.05310>
- [18] D.-A. Clevert, T. Unterthiner, and S. Hochreiter, "Fast and accurate deep network learning by exponential linear units (elus)," 2016. [Online]. Available: <https://arxiv.org/abs/1511.07289>
- [19] D. P. Kingma and J. Ba, "Adam: A method for stochastic optimization," 2017. [Online]. Available: <https://arxiv.org/abs/1412.6980>
- [20] S. Zhong, T. Power, A. Gupta, and P. Mitrano, "PyTorch Kinematics," 2024.
- [21] C. Lugaresi, J. Tang, H. Nash, C. McClanahan, E. Uboweja, M. Hays, F. Zhang, C.-L. Chang, M. Yong, J. Lee, W.-T. Chang, W. Hua, M. Georg, and M. Grundmann, "Mediapipe: A framework for perceiving and processing reality," in *Third Workshop on Computer Vision for AR/VR at IEEE Computer Vision and Pattern Recognition (CVPR) 2019*, 2019.